

# Florida Hurricane Catastrophe Model

## Literature Review and Technical Research Plan

Prepared for internship project

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### 1 Purpose

The objective is not to rebuild a vendor catastrophe model. The objective is to build a transparent miniature catastrophe model that explains the main mechanics: hazard  $\rightarrow$  vulnerability  $\rightarrow$  insured loss  $\rightarrow$  portfolio risk. The model should score hypothetical hurricane exposure sets, produce policy-level and portfolio-level expected losses, quantify uncertainty, and remain explainable to an ILS team.

Available data:

- **claims:** `hurr_id`, `property_id`, `loss`.
- **exposure:** event-property rows with location, construction, occupancy, year built, replacement value, deductible, stories, gross area.
- **hurricane\_data:** storm-track snapshots with time, position, wind, pressure, Saffir-Simpson category, RMW, storm speed/direction, and quadrant wind radii.

### 2 Selective literature review

This is the short list that is worth using. More papers would dilute the project.

#### Core references already provided

**Grossi and TeHennepe (2008), Catastrophe Modelling Fundamentals.** This gives the project architecture: stochastic event module, hazard module, vulnerability module, and financial analysis module. It also gives the portfolio outputs that matter in insurance and ILS: AAL, exceedance probability curves, return periods, coefficient of variation, and technical pricing loads. Use this as the conceptual spine.

**Grant and Atkinson (2023), Residential Wind Loss Mitigation Case Study.** This is directly relevant because it uses Florida residential claims and construction variables. The important lesson is negative: building variables alone have low explanatory power unless wind-field and geography are included. The paper also supports percent loss, loss/coverage, as a natural target and gives benchmark signs: hip roofs reduce losses, newer/post-code construction reduces losses, and geography near the storm path matters.

#### Additional references worth adding

**Powell et al. (2005), State of Florida hurricane loss projection model.** Use for the hazard module. It motivates translating storm-track snapshots into local wind intensity at each exposure, not merely using storm-level maximum wind. This is the key upgrade over a naive regression.

**Pinelli et al. (2011), Application to Florida Public Hurricane Loss Model.** Use for the vulnerability module. It shows how the Florida public model links wind hazard to residential structural vulnerability and insured loss. The value is not to reproduce its engineering detail, but to copy the modular logic and explain that vulnerability should be conditional on both hazard intensity and property attributes.

**Pita et al. (2013/2015), hurricane vulnerability estimation review.** Use only to justify method choice. It distinguishes empirical, engineering, and hybrid vulnerability approaches. Your data support a hybrid empirical model: physical hazard features from track data plus statistical vulnerability estimated from claims.

**Landsea and Franklin (2013), HURDAT2 uncertainty.** Use for limitations and robustness. Best-track positions, intensities, and wind radii are estimates, not ground truth. This justifies sensitivity tests around wind and distance features and prevents overclaiming precision.

**Gulati et al. (2017), PML for the Florida Public Hurricane Loss Model.** Use for portfolio risk framing. It motivates reporting not only mean loss but also tail metrics: event loss distribution, exceedance curves, return-period losses, and portfolio concentration.

### 3 Recommended model

#### 3.1 Core choice

Use a **hybrid two-part physical-statistical model**. It is simple enough to explain, but materially better than a flat regression because it respects the catastrophe model structure.

For event  $i$ , property  $j$ :

$$\begin{aligned} \text{storm track} + \text{property location} &\rightarrow h_{ij} \\ h_{ij} + x_{ij} &\rightarrow \text{damage or claim response} \\ \text{damage} + \text{deductible/replacement value} &\rightarrow \text{insured loss} \end{aligned}$$

where  $h_{ij}$  are hazard features and  $x_{ij}$  are exposure features.

The two-part structure is preferable because hurricane claims are likely zero-inflated: many exposed policies have zero loss, while positive claims are skewed.

#### 3.2 Training table

Start from the full exposure table. Left join claims by (`hurr_id`, `property_id`):

$$L_{ij} = \begin{cases} \text{loss}_{ij}, & \text{if claim row exists,} \\ 0, & \text{otherwise.} \end{cases}$$

Define replacement value  $V_{ij}$  and deductible  $D_{ij}$ . Use the loss ratio

$$Y_{ij} = \frac{L_{ij}}{V_{ij}}, \quad Z_{ij} = \mathbf{1}\{L_{ij} > 0\}.$$

Winsorise or cap extreme  $Y_{ij}$  values only if they are data errors or impossible ratios. Otherwise keep tail information and use robust diagnostics.

#### 3.3 Hazard feature engineering

For each property-event row, compute features from all storm snapshots for that hurricane.

Let property coordinates be  $(\phi_j, \lambda_j)$  and storm snapshot  $t$  be  $(\phi_{it}, \lambda_{it})$  with wind  $W_{it}$ , pressure  $P_{it}$ , RMW  $R_{it}$ , direction  $\theta_{it}$ , and quadrant radii  $R34, R50, R64$ .

Distance:

$$d_{ijt} = \text{haversine}\{(\phi_j, \lambda_j), (\phi_{it}, \lambda_{it})\}.$$

Closest approach:

$$d_{ij}^{\min} = \min_t d_{ijt}.$$

Storm-side indicator. Let  $u_{it}$  be the storm motion vector and  $r_{ijt}$  the vector from storm centre to property. Then

$$\text{side}_{ijt} = \text{sign}(u_{it,x}r_{ijt,y} - u_{it,y}r_{ijt,x}).$$

Right-of-track is usually more damaging in Northern Hemisphere storms because translational speed adds to rotational winds.

Simple local wind proxy. Avoid a complex parametric wind-field model unless time allows. A defensible proxy is:

$$\hat{w}_{ij} = \max_t W_{it} g\left(\frac{d_{ijt}}{R_{it}^q + \epsilon}\right),$$

where  $R_{it}^q$  is the relevant quadrant wind radius and

$$g(a) = \exp(-\alpha a), \quad \alpha > 0.$$

This preserves monotonicity: higher storm wind increases hazard; greater distance reduces hazard; larger wind field increases affected area. Tune  $\alpha$  by validation, or use fixed values and sensitivity tests.

Minimal hazard feature set:

$$H_{ij} = \{\hat{w}_{ij}, d_{ij}^{\min}, \min_t \text{LANDFALL}_{it}, \max_t \text{USA\_SSHS}_{it}, \min_t P_{it}, \max_t \text{RMW}_{it}, \text{right-of-track}\}.$$

### 3.4 Exposure features

Use variables that are present and explainable:

$$X_{ij} = \{\log V_{ij}, D_{ij}/V_{ij}, \log(1 + \text{gross\_area}), \text{stories}, \text{age}, \text{construction}, \text{occupancy}\}.$$

where

$$\text{age} = \text{event\_year} - \text{year\_built}.$$

If year built is missing, include a missingness flag. Do not silently drop large exposure blocks.

### 3.5 Two-part loss model

Part 1: claim occurrence.

$$Z_{ij} \sim \text{Bernoulli}(p_{ij}),$$

$$\text{logit}(p_{ij}) = \beta_0 + \beta_H^\top H_{ij} + \beta_X^\top X_{ij} + \beta_{HX}^\top (H_{ij} \otimes X_{ij}).$$

Keep interactions sparse. The most useful interactions are local wind with construction, local wind with age, local wind with stories, and local wind with deductible ratio.

Part 2: positive severity. For  $L_{ij} > 0$ ,

$$S_{ij} = \log \left( 1 + \frac{L_{ij}}{V_{ij}} \right),$$

$$S_{ij} = \gamma_0 + \gamma_H^\top H_{ij} + \gamma_X^\top X_{ij} + \gamma_{HX}^\top (H_{ij} \otimes X_{ij}) + \varepsilon_{ij}.$$

Use Gamma GLM or lognormal regression if residuals are skewed. The default should be lognormal severity because it is easy to explain.

Expected ground-up loss:

$$\mathbb{E}[L_{ij}^{GU}] = V_{ij} p_{ij} \mathbb{E} \left[ \frac{L_{ij}}{V_{ij}} \mid L_{ij} > 0 \right].$$

If modelling insured loss after deductible for a new exposure, apply the financial layer:

$$L_{ij}^{INS} = \max(L_{ij}^{GU} - D_{ij}, 0).$$

If the historical claims are already paid insured losses, state this and use the deductible mainly as an explanatory feature rather than re-applying it to historical losses.

### 3.6 Portfolio aggregation

For event  $i$ , predicted event loss is

$$\hat{L}_i = \sum_j \hat{L}_{ij}^{INS}.$$

If event probabilities  $p_i$  are available or can be assumed equal across a synthetic event set, then

$$AAL = \sum_i p_i \hat{L}_i, \quad \sigma_L = \sqrt{\sum_i p_i (\hat{L}_i - AAL)^2}, \quad CV = \frac{\sigma_L}{AAL}.$$

The exceedance probability curve is

$$EP(\ell) = \mathbb{P}(L > \ell) = \sum_i p_i \mathbf{1}\{\hat{L}_i > \ell\}.$$

Return-period loss at return period  $T$  is the  $\ell$  satisfying

$$EP(\ell) = \frac{1}{T}.$$

A simple technical price can be reported as

$$\text{Price} = 1.15 \times AAL + 0.20 \times CV \times AAL,$$

clearly labelled as illustrative rather than market pricing.

## 4 Validation and robustness

**Validation split.** Use leave-one-hurricane-out validation. Random row splits leak storm information because the same hurricane appears in both train and test.

**Policy-level metrics.** AUC/log-loss for occurrence; RMSE/MAE for positive loss ratio; calibration plots by predicted-risk bucket.

**Event-level metrics.** Compare actual versus predicted aggregate event loss by hurricane. This is more relevant for ILS than only row-level fit.

**Portfolio metrics.** Compare AAL, CV, EP curve, 1-in-50, 1-in-100, and 1-in-250 event losses under alternative specifications.

**Robustness tests.**

- Baseline: exposure-only model.
- Main model: exposure plus engineered hazard features.
- Specification test: lognormal severity versus Gamma severity.
- Hazard sensitivity: change distance-decay parameter  $\alpha$ .
- Data sensitivity: remove extreme losses; compare results with/without cap.
- Spatial sensitivity: add county/sub-area fixed effects if they improve validation but explain the risk of overfitting geography.

## 5 Why this model should stand out

A plain regression on construction, age, and replacement value ignores the main catastrophe driver: where the storm went and how intense it was near each property. A full engineering model is not feasible with the available columns. The proposed hybrid model is the right middle point: it follows vendor-model logic, uses the available data, is defensible mathematically, and can be explained in a presentation without hiding behind a black box.

## 6 References

1. Grossi, P. and TeHennepe, C. (2008). *Catastrophe Modelling Fundamentals*. The Review - Worldwide Reinsurance.
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5. Pita, G. et al. (2013/2015). *State of the art of hurricane vulnerability estimation methods: a review*.
6. Landsea, C. W. and Franklin, J. L. (2013). *Atlantic Hurricane Database Uncertainty and Presentation of a New Database Format*. Monthly Weather Review.
7. Gulati, S. et al. (2017). *Probable Maximum Loss for the Florida Public Hurricane Loss Model*. ASCE.